

## POWER AND RFD TRAINING

Before looking at how to train power, it's important to understand what power is. In physics, the following definition is used:

$$\text{Power} = \text{Force} \times \text{Velocity}$$

Being dependent on two factors, if one factor has a low value, the end result (power) will be lower than if both factors are of a high value. Therefore, power depends on a climber's level of strength. Greater strength means greater power. If you don't train maximum strength, you'll have less power. Easy. Simple. Concise. As for speed, this is determined by:

- the amount of type II fibres in your muscles (which can be altered slightly via neural adaptation to max strength training but is largely genetic and therefore out of your control); and
  - the volitional action of lifting a weight/doing a move at maximum speed, although this is conditioned by a).
- Some coaches recommend slow pull-ups (both the upward/concentric phase and the downward/eccentric phase).

However, a study by Earp *et al.* (2016) on the effects of lifting a load at different speeds found:

Greater muscle activation and greater RFD (rate of force development) at maximum speed.

Greater tendon compression at the start and the end of the movement.

Greater tendon degradation/damage at slower speeds.

These results show that working at maximum speed not only improves performance but also helps prevent tendon injury.

In relation to improving RFD, in addition to their work on movement at maximum voluntary concentric speed, Levernier and Laffaye (2019) studied 14 climbers on the French national team who climbed at an elite level, competed in world-ranking competitions and climbed above F8b. The study measured maximum grip strength and RFD in open-hand, half-crimp and full-crimp positions. The climbers were split into two groups and both trained for four weeks. The control group followed a normal bouldering training programme, while the experimental group did one-arm hangs (120-degree lock-offs), three times a week, on an edge of their choice between 6 and 25 millimetres deep, on which they could hang for six seconds before failure (falling off).

### Number of sets: 2

### Number of hangs per set: 6

### Rest between sets: 3 minutes

|                  |                             |
|------------------|-----------------------------|
| Open-hand left   | Length of hang: 4–6 seconds |
| Half-crimp left  | Length of hang: 4–6 seconds |
| Half-crimp left  | Length of hang: 4–6 seconds |
| Open-hand right  | Length of hang: 4–6 seconds |
| Half-crimp right | Length of hang: 4–6 seconds |
| Half-crimp right | Length of hang: 4–6 seconds |

The results showed no gains in maximum grip strength but considerable improvement in RFD<sub>200ms</sub>, which increased by 32 per cent in the open-hand position, 27.5 per cent in half crimp and 28 per cent in full crimp, despite not even training in this position.

The authors believe that these incredible results (bearing in mind that even the smallest gains are a huge achievement for elite climbers) were down to neural adaptation, an increased number of electrical impulses transmitted from motor neurons to muscle fibres, which occurs at the start of a muscle contraction and is decisive in the first 100 to 200 milliseconds.

## CAMPUS BOARD TRAINING

The campus board was invented in 1988 by legendary climber Wolfgang Güllich to build the power he needed to climb *Action Directe*, the world's first F9a.

Campus board training is therefore a tool for developing a specific quality: power and RFD. Using it for anything else or using it incorrectly will at best lead to inefficient training and at worst to an unacceptably high risk of injury.

To generate power when campusing, you need both strength (to hold on to the rungs/holds) and speed (to move from one rung to the next). So, if you're not moving at speed (we've all seen a beginner or someone who's very fatigued fighting to do one more move, constantly readjusting their grip), there's no power, and if there's no power then you're not using the campus board efficiently.

If you can't do each move at maximum speed, you should consider whether this type of training is really that suitable. For example:

If you have to readjust your grip on each move (a sign that you aren't able to hold the rungs), you need to work on maximum finger strength instead by using larger rungs and doing more deadhanging.

If you can't move from one rung to the next (a sign that you aren't able to lift your own body weight), you need to work on maximum pulling strength by doing pull-ups. Using either velocity-based training or the maximum intensity methods, try training until you can do at least the same number of pull-ups as there are rungs on your campus board.

Without sufficient strength or speed, the pulleys and tendons in your fingers (which aren't prepared for such levels of stress) will be exposed to exponentially more time under tension, and the risk of injury will skyrocket. Training should improve performance and prevent injury, so if what you're doing is likely to injure you instead, you're probably doing something wrong!

If you have the grip strength and pulling strength to use the campus board safely but you lack the coordination to campus properly, I'd recommend campusing with feet on while you learn the movement.

A practical recommendation for increasing power and RFD would be to do half the maximum possible number of moves (for example, EL 5(10)) or to stop as soon as you're moving 15 to 20 per cent slower than the first rep of the set (González-Badillo and Rivas, 2002, within Balsalobre-Fernández and Jiménez-Reyes, 2014). First set: start campusing up the board. Even if you don't make it to the top, end your set as soon as you start to slow down (even if you've only done one or two moves). If you get to the top of the board without slowing down, drop off and do another lap. Repeat until you start to slow down. With this method, you could have a set of 10 moves, 6 moves, 7 moves, 4 moves, or so on.

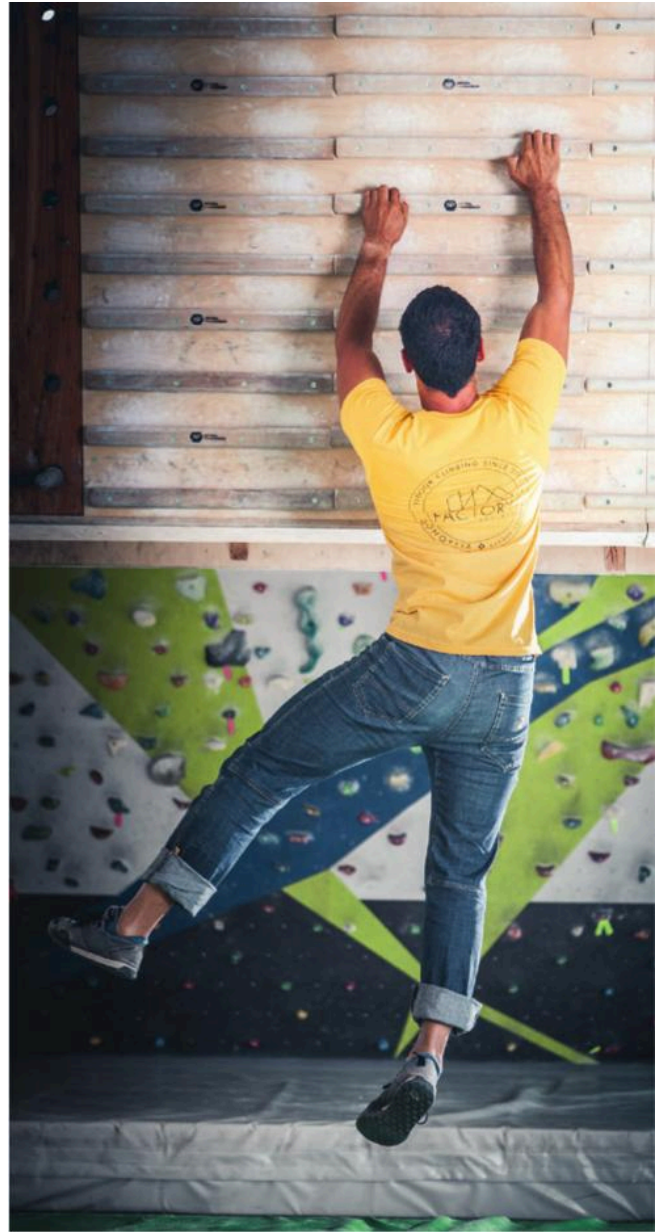
How many sets should you do? Six? Ten? Two? How many moves per set? The answer is simple: it depends on the individual. When planning this type of training, remember that the campus board is a tool which has been designed to train power and RFD. Therefore, as soon as your speed drops, you're no longer working power. If you keep going when you can't move fast enough, you'll create unnecessary fatigue, make it harder to achieve your training goals and increase your risk of injury. In short: a tremendously inefficient, illogical way to train.

If you're going to campus back down, remember that any type of eccentric muscle movement produces greater force than concentric movements. What's more, this eccentric movement will produce even greater force due to the acceleration caused by gravity. As such, I strongly recommend using an open-grip position as your tendons and pulleys will have to bear whatever extra force your muscles are unable to absorb or dissipate. Again, it comes down to common sense and an honest, critical approach. Ask yourself: are my muscles and soft tissue really ready for the stress of campusing back down? If I campus down instead of dropping off, will I be able to campus back up again with enough speed? And will this take too much out of me, meaning I won't be able to go as fast in the next set (cutting short my training for power)?



BUMPS

1



BUMPS

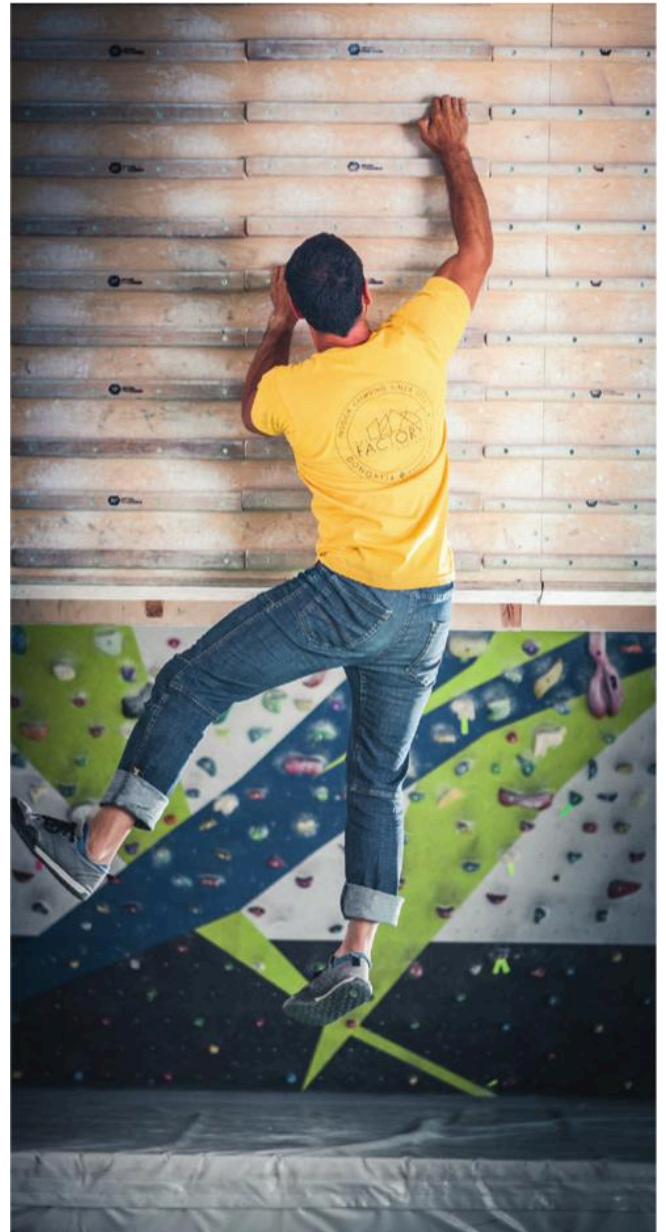
2





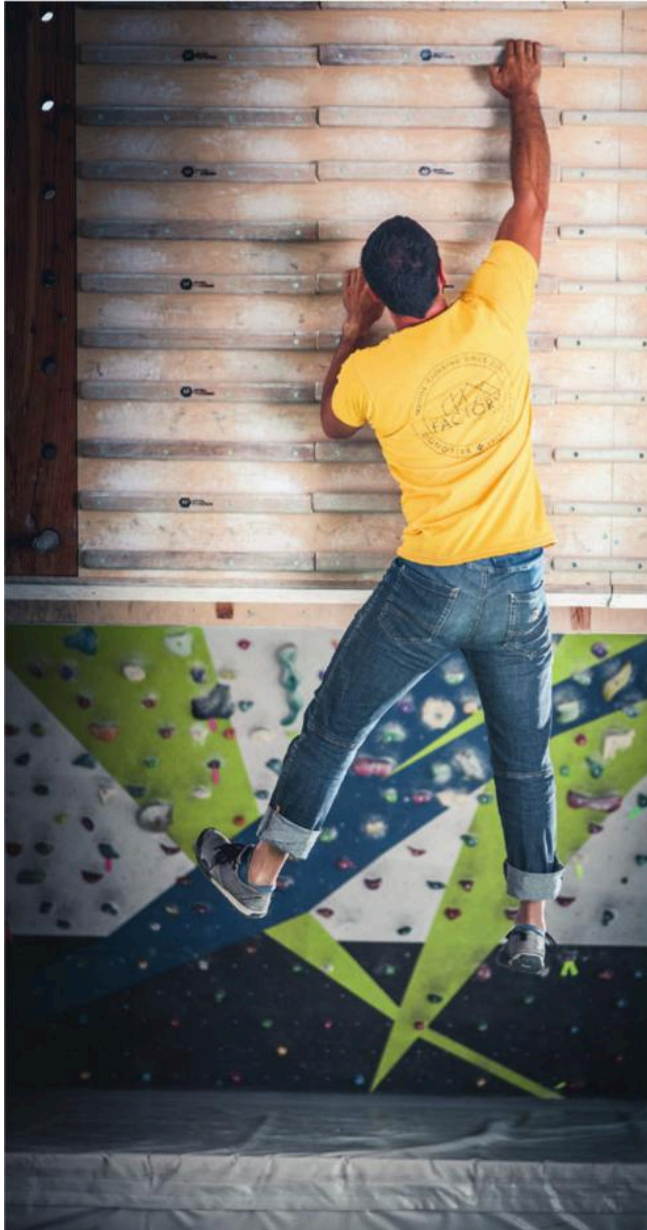
BUMPS

3



BUMPS

4



BUMPS

Campus board training is a perfect example of the ‘no pain, more gain’ mantra or my personal training philosophy: ‘don’t train more: train better’.

As for campusing with added weight, you should ask yourself two questions: Will I be able to maintain a high degree of power and RFD? Campusing with added weight means you’ll need to apply more force, but if you lose speed then you lose power, in which case the extra weight is counterproductive. You should only add weight if you can maintain the same speed and it’s not going to jeopardise RFD.

Can my tendons cope with this added weight? I’ve said it a lot, but the aim of this book is to encourage the most efficient way of training. A training-related injury could stop you from training or climbing for quite some time, which would obviously have a big impact on efficiency. The possibility of injury should never, under any circumstances, enter into your training programme.

I’d recommend the following progression of campus board exercises, listed from least to most powerful (Matros *et al.*, 2013):

*Bumps:* with two hands on the starting rung, bump one hand up rung by rung without moving the bottom hand. This is a great exercise for learning how to train RFD on the campus board.

*Matching ladders:* with two hands on the starting rung, campus up the board matching hands on each rung. Alternate the leading hand if you want to work symmetrically or lead with the same hand if you want to work one side first and then the other.

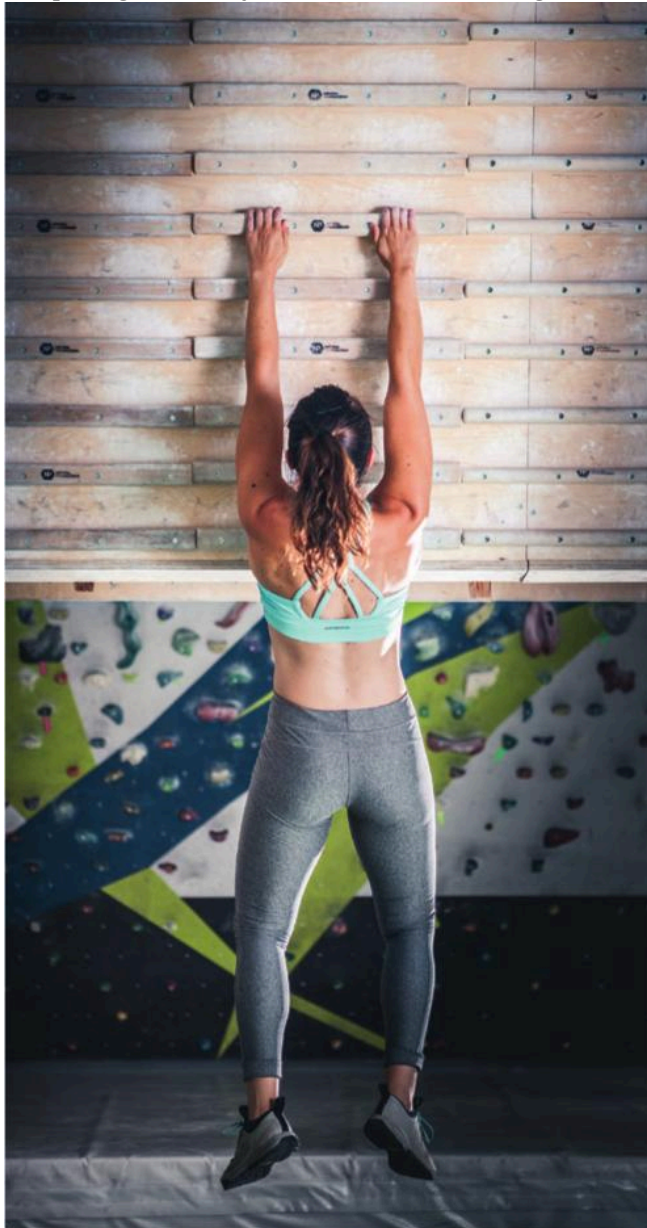
*Standard ladders:* with two hands on the starting rung, campus one hand on even rungs and one hand on odd rungs. This makes the moves slightly bigger than with matching ladders, upping the intensity of the exercise (requiring



more power). See [images on following pages](#).

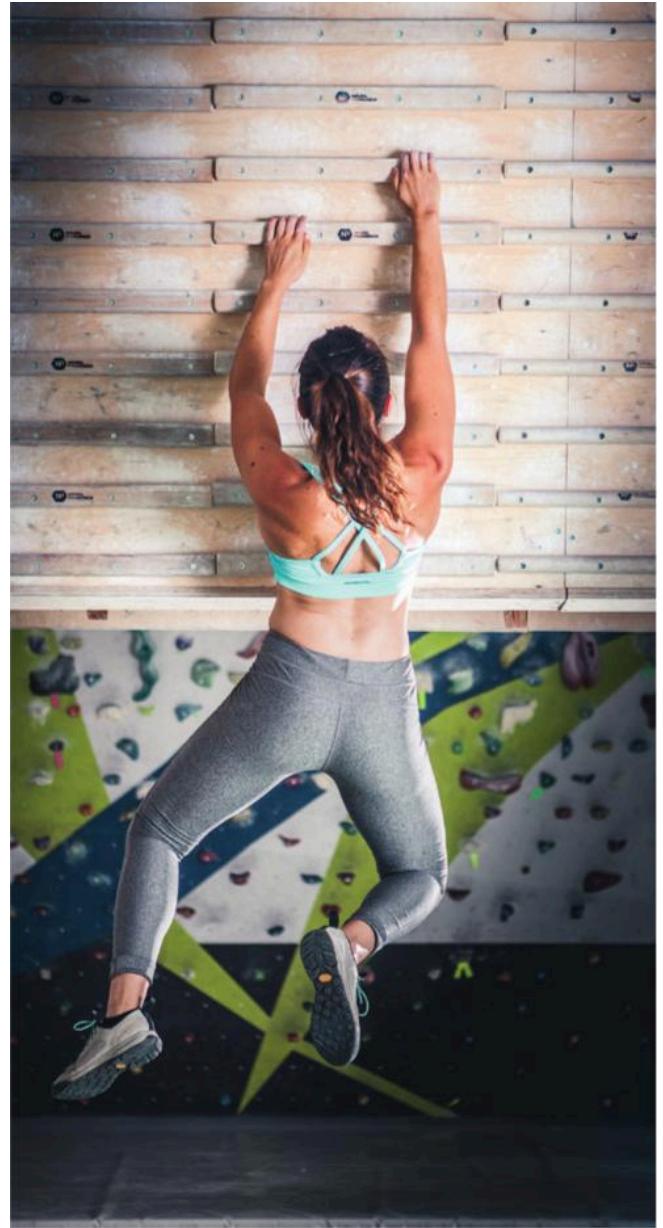
*Skipping rungs:* as with ladders, you can do this with or without matching. Not matching hands requires considerably more strength and speed.

*Maximum extension:* the power exercise par excellence. From your starting rung, campus to the furthest possible rung. Depending on the size of the campus board and your level of power, you can then repeat with the other hand, campusing as far as you can without matching hands.



MATCHING LADDERS

1



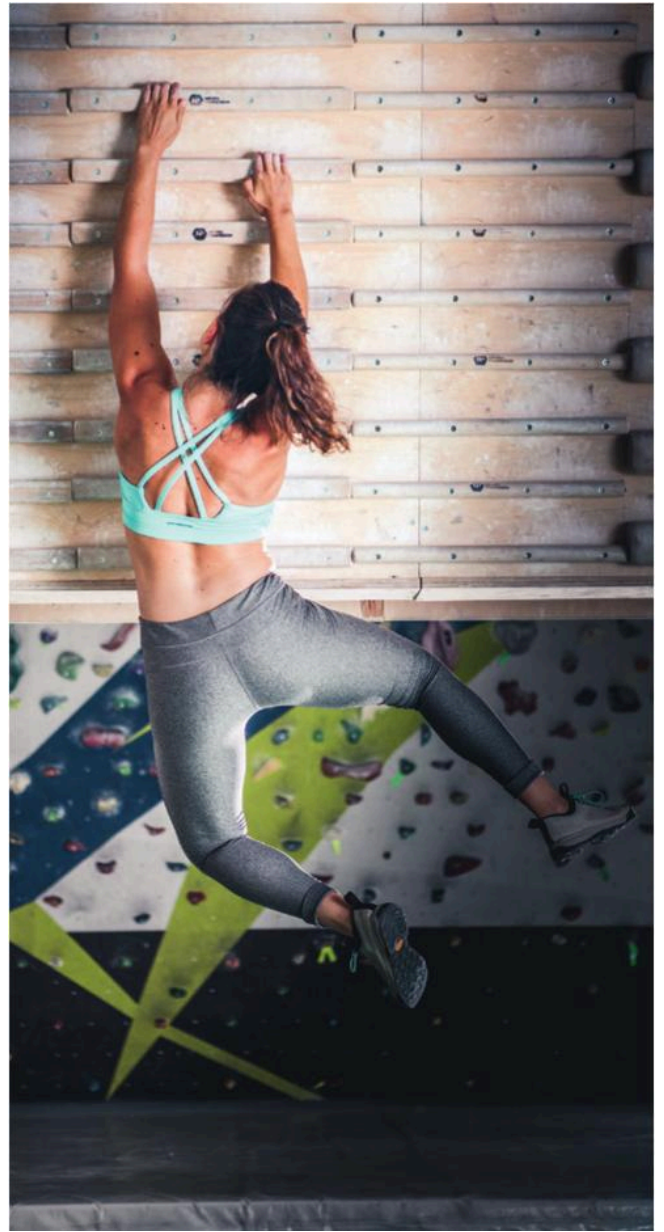
MATCHING LADDERS

2



MATCHING LADDERS

3



MATCHING LADDERS

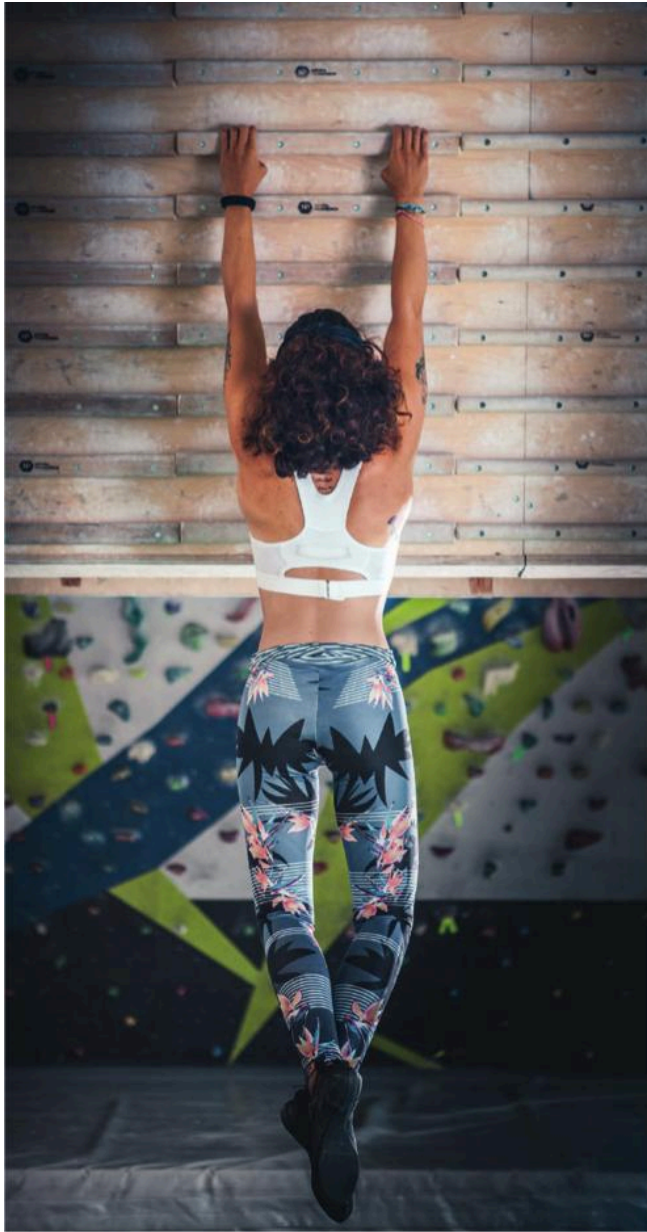
4





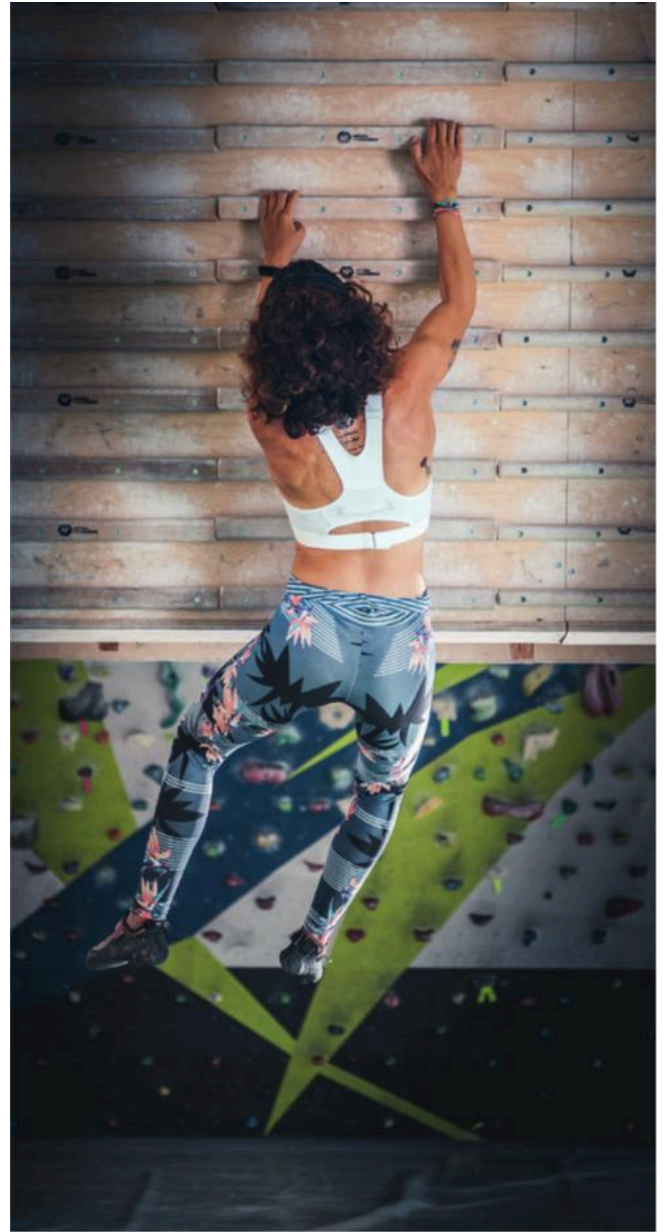
MATCHING LADDERS





STANDARD LADDERS

1



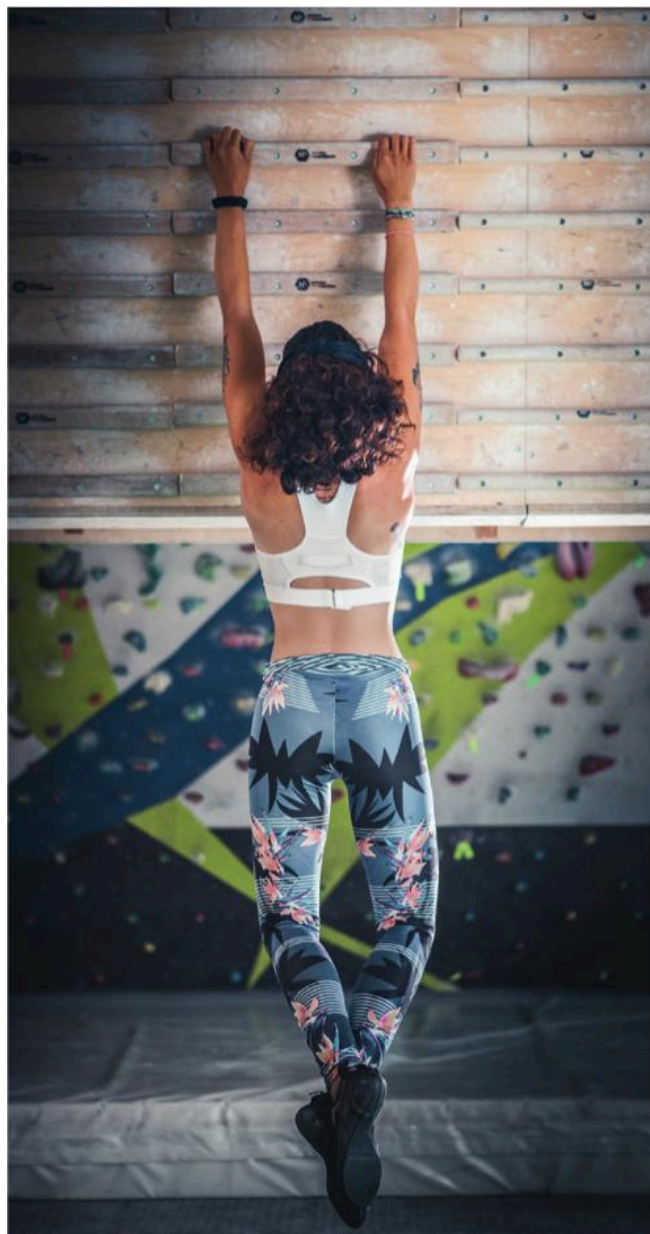
STANDARD LADDERS

2



STANDARD LADDERS

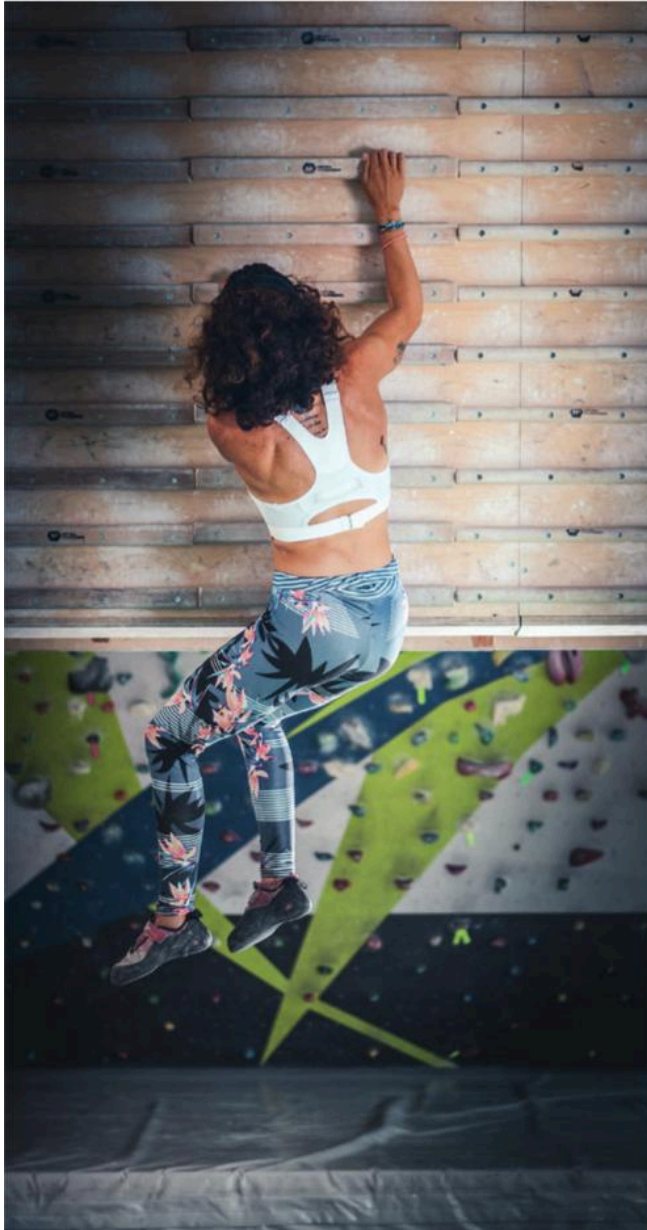
3



SKIPPING RUNGS (WITHOUT MATCHING)

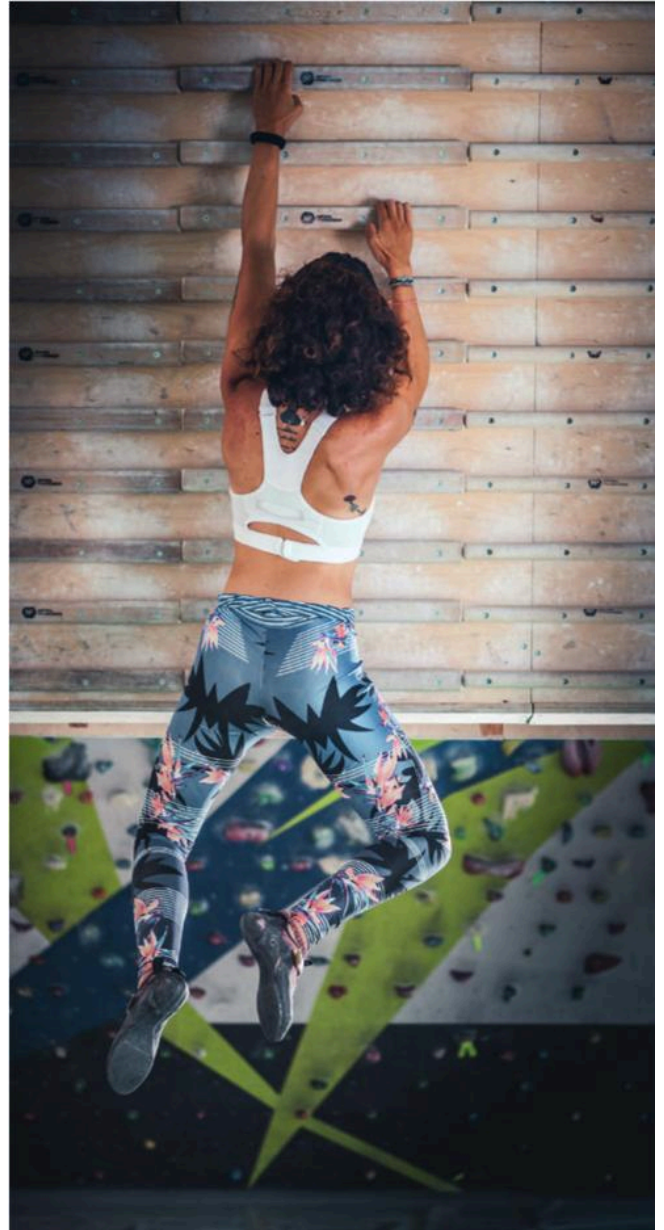
1





SKIPPING RUNGS (WITHOUT MATCHING)

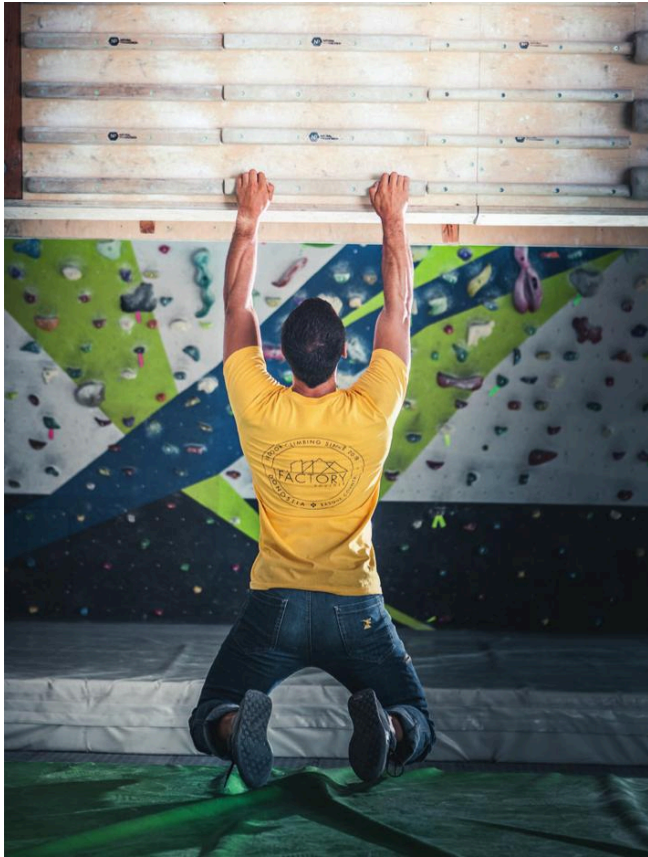
2



SKIPPING RUNGS (WITHOUT MATCHING)

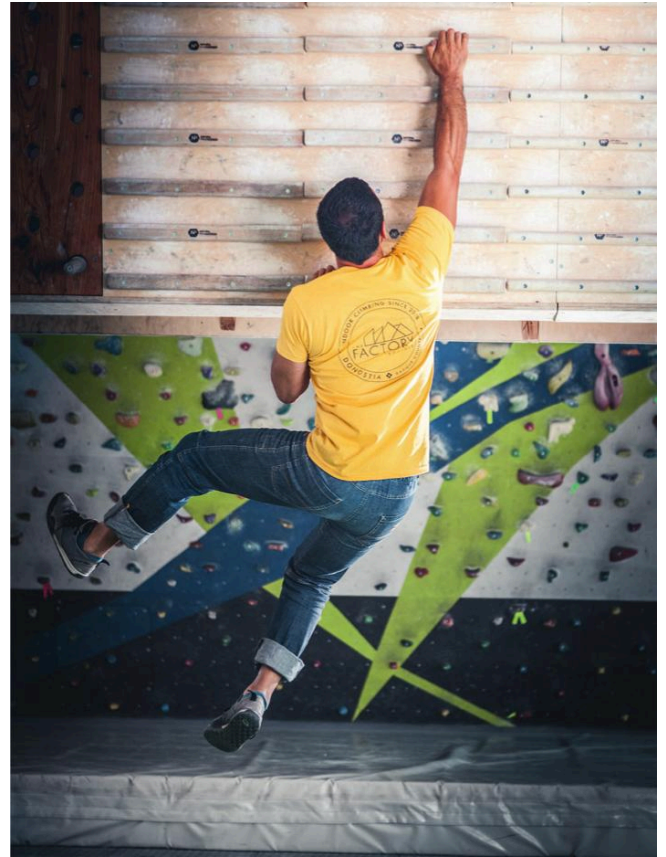
3





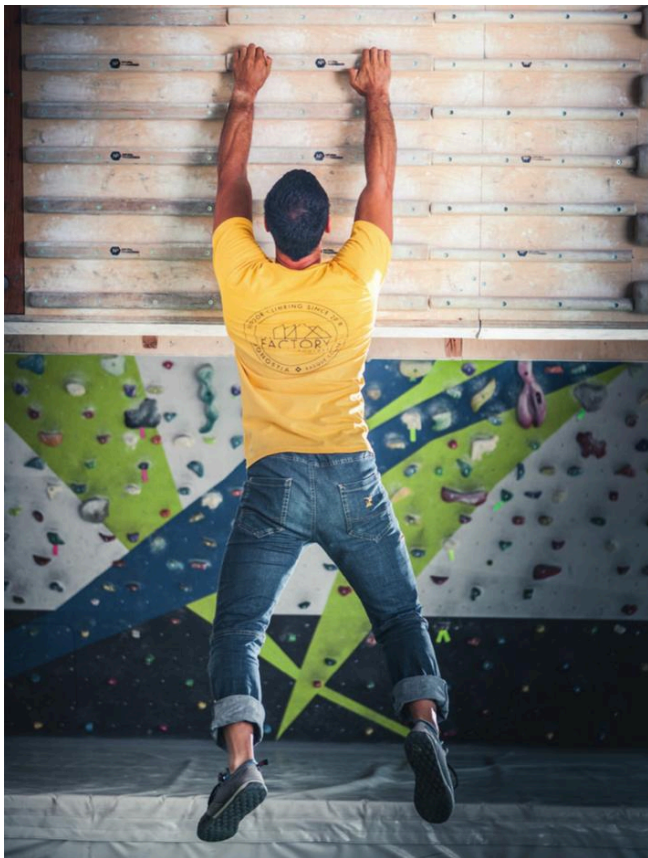
MAXIMUM EXTENSION

1



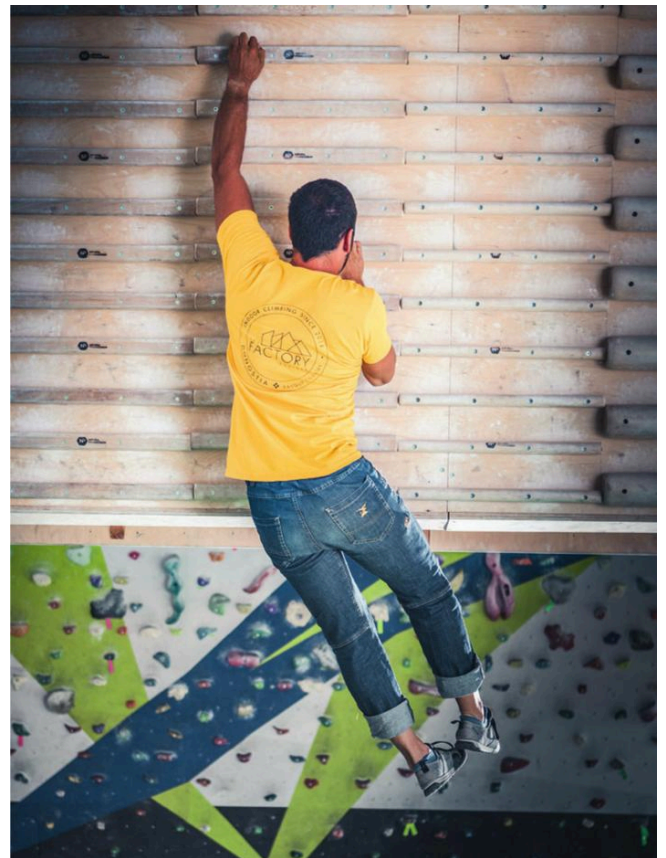
MAXIMUM EXTENSION

2



MAXIMUM EXTENSION

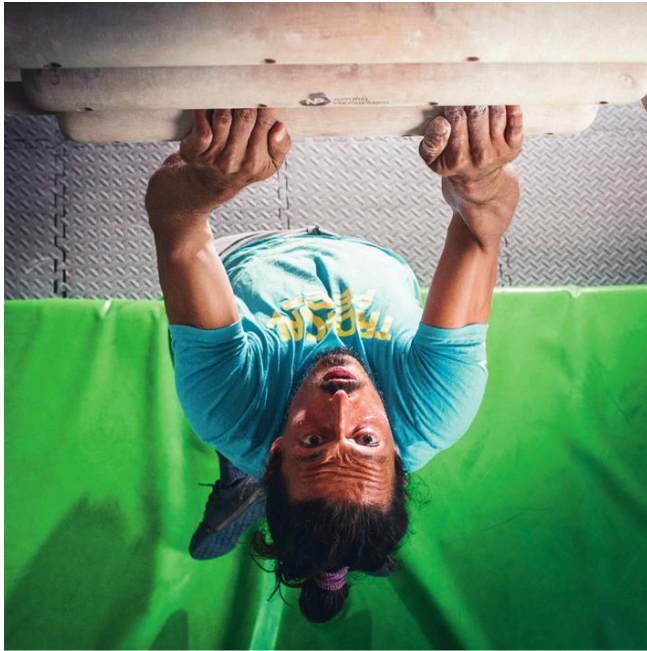
3



MAXIMUM EXTENSION

4





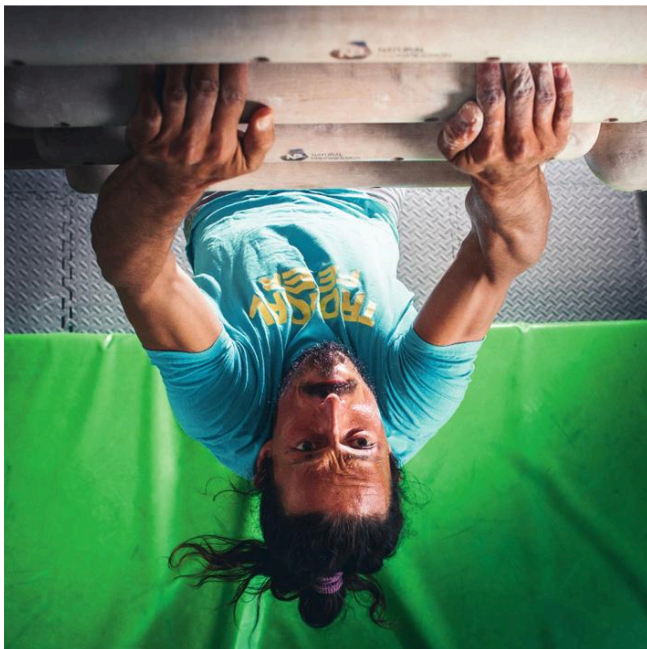
DOUBLE DYNOS

1



DOUBLE DYNOS

2



DOUBLE DYNOS

3

With the trailing/lower arm, you could try the following progression to increase range, power and speed (providing you can still maintain RFD and therefore continue the exercise): start by locking off at shoulder height, then progress to a deeper lock-off at rib height, waist, hip, and so on. Extremely strong climbers could even lock off with the arm fully extended below, much like a muscle-up.

*Double dynos:* campus with both hands at the same time, either rung by rung or skipping rungs (depending on ability). This exercise requires not only a huge amount of power but also really good motor skills and hand-eye coordination.

At what point should I let go to go for the next rung? To answer this question, let's look at some basic physics: when an object is launched vertically into the air, it will gradually slow down, due to gravitational pull, until its velocity is equal to zero. At this point, it changes direction and starts falling back down. The object has a constant or uniform rate of acceleration. Any exercise on the campus board starts with a pulling motion (a pull-up). A pull-up starts with the arms fully extended and ends when the hands are in line with the shoulders. The movement starts with the application of force, followed by an inevitable loss of velocity at the end of the pull. In other words, from zero velocity (a static hang), you'd apply a maximum amount of force, experience a gradual loss of velocity and at the end of the movement you'd return to zero velocity. It is at this point of zero velocity when you should let go with

your leading hand and go for the next rung. Why? Because each time you campus, you lead with either your left or your right hand, and the direction of travel is slightly different on each side. As such, you need to change direction with each move and the point of zero velocity is when you need the least amount of force to do this. What's more, the body is weightless at this point as gravity still hasn't begun to exert any downward force. In summary: go for the next rung at the high point of your pull (hands in line with shoulders) before gravity kicks in and starts pulling you back down.

## RECOVERY TIME DURING AND BETWEEN SESSIONS

As with maximum strength training, to train efficiently, it's important to keep fatigue to a minimum to ensure the right level of intensity and the greatest possible production of force and speed.

Once again, there should be no more than a 15 to 20 per cent velocity loss between reps, which is the same as doing half the maximum number of moves that you're able to do. However, as less force is required (since in most cases you're 'only' working with your own body weight and not adding extra weight), the intensity is maintained by doing more repetitions than in maximum strength training. This causes a higher degree of mechanical fatigue, resulting in higher levels of hydrogen and ammonium.

The amount of rest between sets and training sessions is very similar to the recommended rest for maximum strength training: three to five minutes between sets to replenish PCr stores, and a minimum of 24 hours between sessions for central nervous system recovery. Due to the higher degree of mechanical fatigue, I'd recommend resting for 7 to 10 minutes after campus board training before moving on to other types of training (bouldering/routes).

## INTEGRATED STRENGTH, POWER AND RFD TRAINING: BOULDERING

Bouldering, aside from being a sport in its own right, is a brilliant way to train for other types of climbing as, generally speaking, it requires greater strength, power and RFD. What's more, boulderers show a greater degree of neural adaptation to building strength, power and RFD (Fryer *et al.*, 2017).



Boulderers also have a higher percentage of type II muscle fibres (stronger and more explosive) than sport climbers, who have more type I fibres (less explosive but more resistant to fatigue) (Laffaye *et al.*, 2016). This gives an interesting insight into compensatory hypertrophy or the selective conversion of type Ix fibres into type II or type



I depending on our performance needs (Benito, 2008). For a sport climber with a good level of endurance but poor maximum strength, power and/or RFD, bouldering will encourage the conversion of type IIx fibres into type II fibres, improving these qualities via neural adaptation and helping the climber to address these weaknesses.

Since the aim of bouldering as a form of training is to increase strength, power and RFD, it's important to remember that:

*Regarding intensity:* you should be training on maximum intensity problems; the limiting factor shouldn't be the number of moves but the explosiveness of the moves (big, dynamic) and the amount of strength (pushing, pulling, lock-off and grip) required to do them. Be wary of very small edges, shallow pockets or monos and remember that injury is not an option!

*Regarding recovery:* you should always rest for three to five minutes between attempts or problems to ensure the best possible recovery. Working at maximum intensity each time you pull on is essential for neural adaptation.

## ENDURANCE TRAINING

As discussed in previous chapters, a good level of endurance in the forearm flexor muscles, specifically the flexor digitorum profundus (FDP), is a defining quality among sport climbers.

Remember that endurance is simply the ability to maintain the production of force. And the more fatigued we become, the harder it is to produce force. This highlights the undeniable link between endurance and efficiency.

We know that a high volume of low-intensity training is essential for aerobic endurance sports (such as rowing, cycling, athletics, cross-country skiing), and it's the same for climbing: hours and hours of indoor and outdoor climbing is essential to build greater efficiency and endurance. In other words, training a couple of days a week at the wall and going to the crag once a month might keep you in shape, but it won't be enough to help you really push your grade.

Therefore, your endurance training should focus both on improving efficiency by doing a high volume of climbing, and on improving the defining aspects of endurance: force-time integral (FTI), PCr stores and the oxidative capacity of your muscles.

What is the best way to target these factors in order to improve them? Scientific research shows that:

Strength training, high-intensity intermittent endurance training and low- to moderate-intensity continuous endurance training improve capillary ratio and density (Alomari *et al.*, 2010, within Philippe *et al.*, 2012).

Improved capillary density has a direct impact on increased blood flow and greater muscle reoxygenation (Robbins *et al.*, 2009; Jensen *et al.*, 2004; Green *et al.*, 1999; McCall *et al.*, 1996; Henriksson, 1992; and Klausen *et al.*, 1981; within Philippe *et al.*, 2012).

Faster reoxygenation means faster replenishment of PCr stores (McMahon and Jenkins, 2002; and Tomlin and Wenger, 2001; within Philippe *et al.*, 2012).

As with maximum strength and power, endurance training can be divided into isolated training (mainly intermittent deadhangs, aka repeaters) and integrated training (different methods of wall-based endurance training). However, the aim of both forms of training is to improve recovery (specifically the recovery of PCr stores). Remember that, in climbing, aerobic/oxidative metabolism supports anaerobic alactic metabolism by producing energy not so we can continue climbing but to bind more phosphate to creatine and provide that burst of energy needed to latch the next hold. As such, optimising all factors that contribute to the aerobic energy system, in order to produce more creatine phosphate and replenish our PCr stores, should be the primary aim of endurance training.

## PHYSIOLOGICAL EFFECTS OF DIFFERENT INTENSITIES

As discussed in previous chapters, the efficiency of the oxidative system (the oxidative capacity of any given muscle) is determined by the blood flow to our muscles and the muscles' ability to use their oxygen stores (deoxygenation) and then replenish them with fresh oxygen (reoxygenation). This section will look at how we can optimise these three factors.

The different intensities at which we train or climb have different sets of demands and also effects on our body. Understanding what determines climbing performance and how these determining factors work (a key aim of this book) is essential for knowing which training method to use, depending on what impact you want to have or what you want to improve.

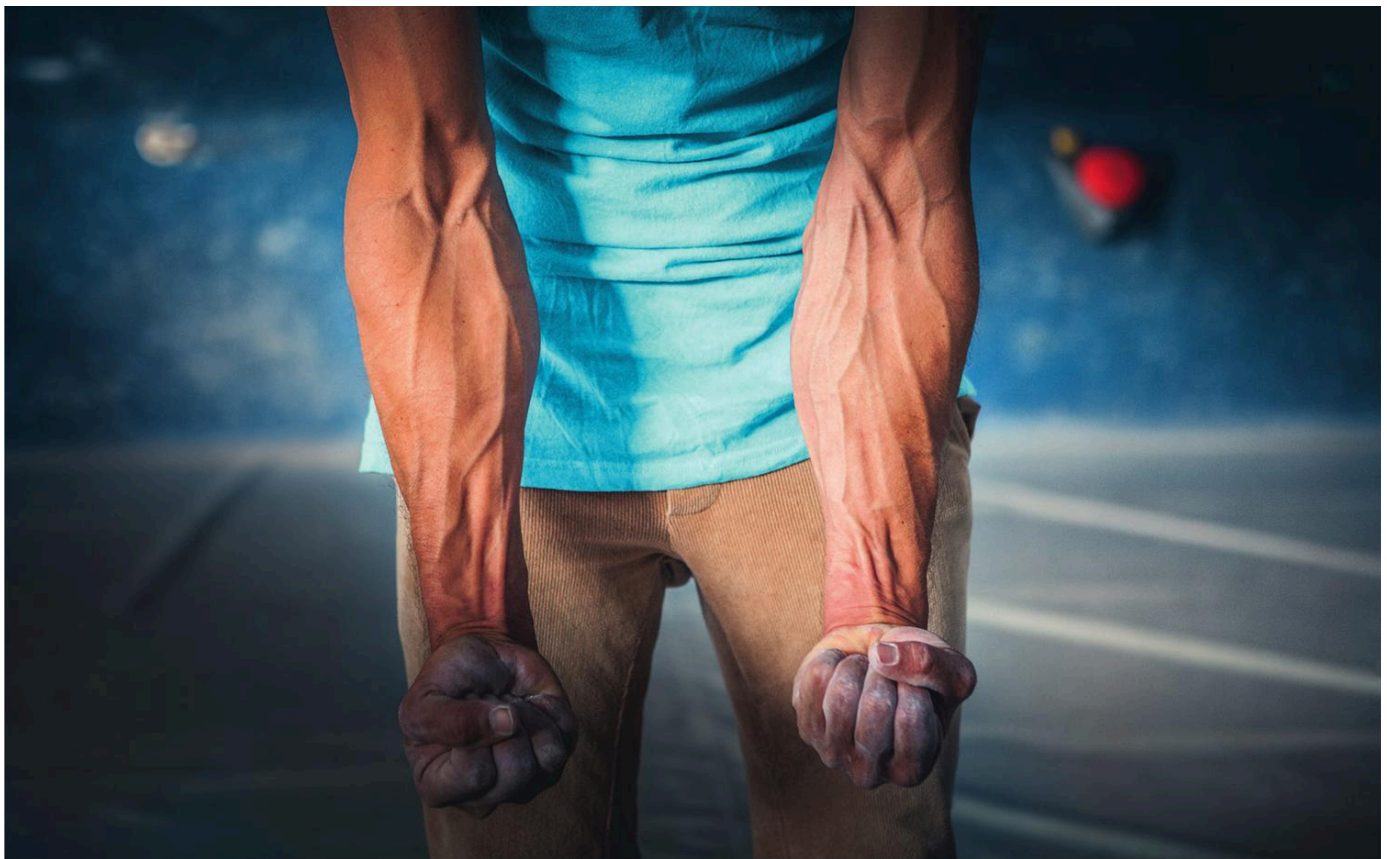
In very simple terms, training can be classified according to:

Intensity: low, medium or high

Form: continuous or intermittent

Different combinations of these two factors will have different physiological effects.

Low-intensity, high-volume training enhances blood flow to the targeted muscles, which increases capillarisation (encouraging angiogenesis, the creation of new blood vessels) by approximately 20 per cent (Mujika, 2012) and improves the reoxygenation capacity of muscle fibres (Baechle and Earle, 2000; López-Rivera, 2014b). In good news, the creation of new blood vessels begins after just one to two weeks of training (Mujika, 2012). In addition to increasing the number of capillaries and, therefore, improving the capillary–muscle fibre ratio, low-intensity, high-volume exercise not only has a direct impact on blood flow but also on muscle fibre. It encourages the creation of new mitochondria in muscle fibre (mitochondrial biogenesis) (Mujika, 2012; House and Johnston, 2014), meaning that each fibre is able to oxidise more fatty acids and produce more energy. However, this process is a little slower than angiogenesis and begins after around six weeks of training (Mujika, 2012). To create new mitochondria, there must be no muscle acidification, meaning no hydrogen produced by lactic acid (House and Johnston, 2014). Interestingly, the absence of metabolic waste, a key feature of low-intensity training, also raises the excitation threshold of the sympathetic nervous system. Certain metabolites ( $H^+$ , lactic acid,  $K^+$ , arachidonic acid, adenosine and diprotonated phosphate) stimulate the sympathetic nerve endings, causing the constriction of blood vessels. However, Mostoufi-Moab *et al.* (1998) found that in non-exhaustive exercise (working at 25 per cent of MVC), these metabolites didn't stimulate the sympathetic nerve endings in the forearm muscles but instead raised the intensity threshold at which these metaboreceptors are activated. In simpler terms, low-intensity training means you'll be able to climb at a higher intensity without constricting the blood vessels in your forearms, meaning you get better blood flow at higher intensities.



At higher intensities, the body starts to favour anaerobic glycolysis over the pairing of anaerobic alactic and aerobic metabolism. But how necessary is it to train or to stimulate this metabolic pathway/anaerobic glycolysis?

A systematic review by Saul *et al.* (2019) of over 350 studies confirmed that the steeper the overhang, the higher the concentration of lactic acid in the blood. That is, if you climb overhanging routes or problems, anaerobic glycolysis will be important to you. But how much lactic acid are we talking? Billat *et al.* (1995) found that after climbing routes graded around F7b, climbers had concentrations of between 4.3 and 5.7 mmol/L in their blood. Similarly, Schöffl *et al.* (2006) found concentrations of  $5.0 \pm 1.3$  mmol/L after a Treadwall test (a rotating training board with variable speeds and angles; like a treadmill but for climbing). So, the answer is ... not very much. Training this metabolic pathway could be useful in preparation for certain sequences or difficult sections on a climb, but that's about it. What's more, you'd also need to be training your ability to flush out metabolic waste while resting. This ability has been linked to a high force–time integral (FTI) (Fryer *et al.*, 2015b), which is a strong indicator of climbing performance. In other words, climbing until your forearms feel like they're going to burst, when you're totally pumped and you can't even close your fingers around a hold, makes little sense if you want to



improve performance. There will be such a build-up of metabolic waste by this point that it will only lead to reduced blood flow to the forearm muscles (due to a build-up of pressure and metaboreceptor stimulation); acidification of muscle cells (making muscle contractions more difficult); and the onset of muscle failure (increasing the risk of injury, increasing the amount of time you need to recover, reducing RFD) – that is to say, the exact opposite of the things you need in climbing.

## **INCREASING FOREARM BLOOD FLOW. CONTINUOUS, LONG-INTERVAL AND INTERMITTENT METHODS. BLOOD FLOW RESTRICTION TRAINING (BFR)**

As we've seen, increasing forearm blood flow is really beneficial when it comes to climbing performance. And to increase it, we need to be training high volume and low intensity. In climbing, intensity is determined by the percentage of maximum strength needed to remain on a handhold and by how long we're holding on for. Then there are additional factors, such as what we're doing with our feet, whether we're clipping, the angle of the wall, and so on, which will vary for each individual climb (López-Rivera, 2014b).

We know that to encourage muscle capillarisation, we need to be training at a low intensity. To be exact, this means under 25 per cent of our maximum grip strength. Why? Because blood flow is greatest at 10 to 25 per cent of our MVC; there's no increase in blood flow at 25 to 40 per cent and blood flow starts to reduce at 40 per cent and over (Barnes, 1980; Byström and Kilbom, 1990; within López-Rivera, 2014b). In addition, some authors consider 25 per cent MVC as the upper limit of aerobic metabolism (Fallentin *et al.*, 1993; Byström, 1994; Kimura *et al.*, 2006; within López-Rivera, 2014b). Finally, the more intense the contraction, the greater the intramuscular pressure, which can occlude the blood vessels. The occlusion of forearm blood vessels begins at 30 per cent MVC, with total occlusion at 50 to 70 per cent MVC (Barcroft and Miller, 1939; Barnes, 1980; Sjøgaard *et al.*, 1998; within López-Rivera, 2014b).

How can you work out the percentage of MVC that you're using? An isometric contraction at 10 per cent MVC can be maintained for one hour; a contraction at 12 per cent MVC can be maintained for around 40 minutes; continuous and intermittent contractions at 25 per cent MVC (10-second contraction/two-second rest) can be maintained for six to eight minutes; and contractions at around 30 per cent MVC can be maintained for about two to four minutes (Rohmert 1960; Byström, 1994; Allison *et al.*, 2004; Frey and Avin, 2010; within López-Rivera, 2014b).

According to López-Rivera (2014b), you can feel whether you're working at a low intensity by checking for the following key signs:

*Forearms:* slight pump and moderate activation, avoiding extreme tightness or pump. Some widening of the blood vessels (heat, flushed skin, more visible veins). Progressive feeling of 'emptiness' and mild pain (between one and two on a subjective one-to-five scale of pump/pain, where five would be the maximum).

*General signs:* slight increase in heart rate and breathing, light perspiration and a progressive awareness, especially towards the end, that you're moving more slowly.

Now we know the best training intensity to encourage capillarisation and to maximise the efficiency of the oxidative energy system, let's take a look at the following protocols proposed by López-Rivera (2014c):

*Continuous method:* 5 to 45 minutes of continuous climbing, staying on the wall the whole time.

*Long-interval method:* two to eight sets of 4 to 20 minutes of continuous climbing, resting for between 45 seconds and three minutes between sets, with a total of 8 to 60 minutes of climbing.

*Intermittent method:* this method is perfect for less advanced climbers who might struggle with the continuous or long-interval methods. Begin by choosing a period of time (for example, eight minutes). Start your timer when you get on the wall. Climb until you reach a level of intensity of over two out of five, then stop climbing and pause your timer. Take as much time as you need to recover, then get back on the wall and restart the timer. Repeat until you've completed the chosen period of time (eight minutes).

What's the recommended volume of climbing? López-Rivera (2014c) suggests aiming to maintain the key physical signs mentioned above for the following periods of time:

From 8 to 12 minutes for beginners.

From 15 to 30 minutes for intermediate climbers.

Over 40 minutes for advanced climbers.

Up to 60 minutes or more for very advanced training or training for multi-pitch routes

Another strategy to consider, borrowed from the world of strength training, is the use of vascular occlusion, blood flow restriction (BFR) or KAATSU training (the original commercial name). In this method, a cuff, band, strap or tourniquet is used to apply pressure at higher than normal systolic levels (above 120mmHg (millimetres of mercury,

the measure of blood pressure)). The cuff is attached at armpit height or above/below the elbow, depending on the target area (whole arm or forearm).

What are the benefits of vascular occlusion? Low-intensity strength training (20 to 40 per cent 1RM) with vascular occlusion has been scientifically proven to give similar results (mainly through sarcoplasmic hypertrophy and therefore not via neural adaptation) to regular high-intensity training (70 to 85 per cent 1RM) (Peña *et al.*, 2013). It's also highly beneficial in injury rehabilitation as you can recover high levels of strength without putting excessive strain on your tendons.

What's more, the effects of BFR training on endurance are also pretty impressive: Counts *et al.* (2016) found a 62 per cent improvement in muscular endurance in the elbow flexor muscles after eight weeks of training with 40 per cent arterial occlusion.

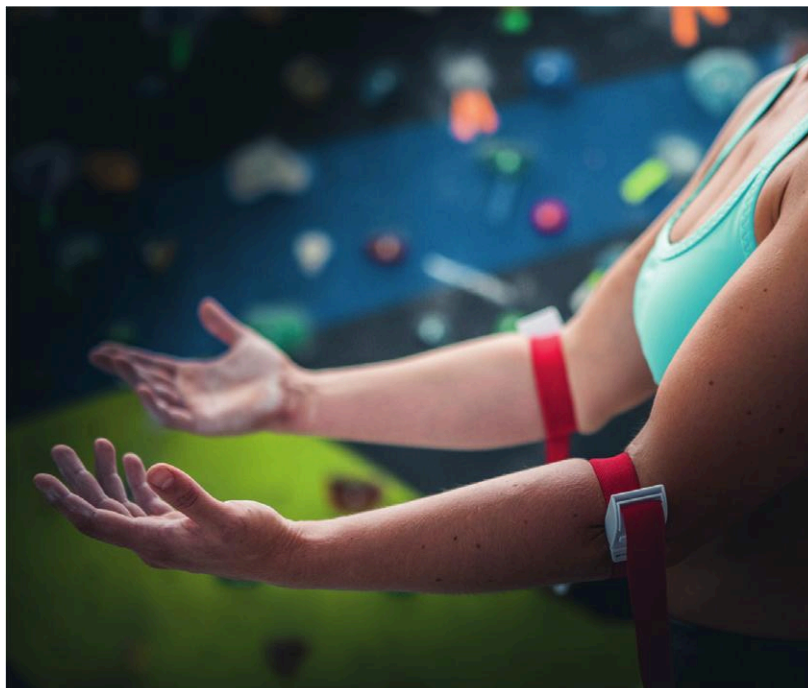
Amani-Shalamzari *et al.* (2019) and Taylor *et al.*, (2016) found that BFR training led to increased capillary density. A very interesting study by Jeffries *et al.* (2018) analysed the effects of preconditioning with BFR on the lower body. For seven consecutive days, while at rest, participants underwent four 5-minute sets of occlusion at 220mmHg on the upper thigh. Seventy-two hours after the seventh day, the participants performed a plantar flexion isometric contraction test (contract for 2.5 seconds, rest for 2.5 seconds) for two minutes at 40 per cent MVC and then at 60 per cent MVC. The study found that muscle reoxygenation was 33 per cent faster at 40 per cent MVC and 43 per cent faster at 60 per cent MVC. These spectacular results suggest that even if we miss a week of climbing for whatever reason (such as travel or work), it's possible to not only maintain but also improve muscle oxidative capacity.

Biazon *et al.* (2019) studied the effects of high-intensity strength training and training at both high and low intensities with BFR. The greatest benefits in terms of deoxygenation were found at low intensities with blood flow restriction.

Horiuchi and Okita (2012) found a significant increase in the secretion of vascular endothelial growth factor (VEGF), which is responsible for angiogenesis (growth of new blood vessels), after using BFR. This is because the production of VEGF is stimulated by hypoxia (a lack of oxygen). What's more, the bioavailability of nitric oxide, a powerful vasodilator that guarantees blood flow to the muscle, increases after the secretion of VEGF. It all sounds pretty good, right?







One of the defining features of BFR is that it promotes the recruitment of type II muscle fibres during low-intensity training. This could be due to the localised hypoxia caused by BFR which, in turn, puts greater demand on the anaerobic lactic energy system to ensure the resynthesis of ATP (Domínguez, 2015).

A basic protocol for BFR training could be summarised as follows (adapted from Peña *et al.*, 2013):

Frequency of training: two to three times a week.

Fifteen moves or more per set, aiming for muscle failure.

Three to five sets per session.

Resting for 30 to 60 seconds between sets, maintaining BFR.

Intensity: low to moderate (20 to 40 per cent). Avoid very small holds or holds that put a lot of strain on the tendons; the aim is mechanical function not muscle tension.

Total time under occlusion: 10 to 15 minutes.

If you're concerned about the safety of this method, the risk of blood clots or other nasty side effects, Manini and Clark (2009) analysed over 30,000 BFR training sessions and found the most common side effects were bruising (13.1 per cent), numbness (1.3 per cent) and slight dizziness (0.3 per cent), while more serious side effects, such as blood clots, only occurred in 0.06 per cent of cases. That said, I wouldn't recommend BFR to anyone at an increased risk of cardiovascular disease.

## **IMPROVING RECOVERY. HIGH-INTENSITY INTERVAL METHODS. ACTIVE WALL-BASED RECOVERY. INTERMITTENT DEADHANGS**

At this stage, it goes without saying that efficient recovery is really important, especially for sport climbers. The question to ask is: how can I improve my recovery? What can I do to recover better?

After a difficult section of climbing, I'm sure many of us would shake out on the next available jug. I hate to say it, but Green and Stannard (2010) have demonstrated that shaking out has no bearing on recovery. They found that in the event of full vascular occlusion, a trained climber becomes comparable to an untrained climber. Moral of the story: you've got to train to avoid getting totally pumped on crux moves, to visualise them and climb them as efficiently as possible, or you'll never be able to climb a whole route at your grade.

A study by Valenzuela *et al.* (2015) sheds some light on how to improve recovery. Participants in their study had three attempts on an indoor F6c route. They had two minutes for each attempt and two minutes to rest between attempts. The study compared the effects of walking and of easy traversing as a means of active recovery. On the one hand, there was no significant difference in blood lactate levels until the final attempt, when levels were lower in climbers who did easy traversing between attempts. On the other hand, loss of hand strength was practically identical for both types of recovery. So, what's the benefit of more active recovery? On their last attempt, the climbers who did easy traversing to actively recover made it higher up the route than the climbers who just did walking.

These findings are especially useful for planning high-intensity interval training. If you want to get better at recovering while on a route, so you can do the crux moves and recover afterwards, it might be better to ditch the typical 4x4 protocol (four sets of four problems in a row at maximum intensity), resting for three minutes between sets, sitting down (or face down!) on the mats, which leaves you totally pumped and drained for the rest of the session. Instead, you could try some low-intensity climbing interspersed with problems of varying lengths at a high or very high intensity, without getting to the point of total vascular occlusion – the point of no return. For example: Two minutes of traversing + a four-move problem at very high intensity + one minute of traversing + an eight-move problem at high intensity + two minutes of traversing + a four-move problem at very high intensity + three minutes of traversing.

A six-move problem at very high intensity + traversing until you feel less pumped + a 10-move problem at high intensity + traversing until you feel less pumped + a four-move problem + traversing until you feel less pumped.

These high-intensity problems or intervals, imitating the crux sections of a route, build strength, power and RFD while mainly using the anaerobic lactic energy system. If we can ensure that these intervals don't leave us 'out of the game', that we don't completely block the blood flow to our muscles, the traverses will help improve our ability to flush out metabolic waste and to reoxygenate our muscle fibres, giving us the best possible chance of getting through the next bit of hard climbing, just like on a route.

Although finding good rests on a route is second nature for more experienced climbers, most beginners never even consider that before getting completely pumped, they could (or might even need to) recover on the route without resting on the rope.

Active recovery is something we need to learn and train. To learn how to recover actively on the wall, I suggest the following progression:

Start climbing on a non-overhanging wall and when you notice a slight pump in your forearms and long before the point of failure (remember, getting totally pumped is no good if you want to climb well afterwards), stop on a good hold (ideally a three-pad hold or a jug which you can get your whole hand around) and take one hand off for a few seconds. Swap hands when the hand you're holding on with starts to fatigue. You could start by swapping hands four to five times on a non-overhanging wall, resting each hand for a minimum of 3 to 10 seconds. When you feel you've recovered, leave the resting hold and carry on climbing. Learning to control your breathing when resting is also an essential part of recovery.

Start climbing on an overhanging wall (upping the intensity) and when you notice a slight pump or fatigue in your forearms, find a good hold or jug on a vertical wall and take one hand off for a few seconds, and then swap to the other. When you feel you've recovered, go back to climbing on the overhanging wall.

Climb on an overhanging wall and recover on the overhanging wall.



REST POSITION (ALTERNATING HANDS)







REST POSITION (ALTERNATING HANDS)

For less-experienced climbers, increasing recovery time will be a real challenge.

For climbers who can already rest and recover while on a route, doing intermittent deadhangs (also known as repeaters) on a fingerboard is the most efficient way to improve recovery. There are numerous protocols, some easy and some hard, some more logical or reliable than others, some that are specific to certain fingerboards and others that will work with any fingerboard. The key is to make sure you can complete the full volume of training for the entire protocol.

Medernach *et al.* (2015) compared the effects of training intermittent deadhangs and training bouldering, in competitors in this discipline. Three 150-minute sessions a week at a density of 1:2 (one day on, two days off) produced a very significant increase in grip endurance on crimps, pinches and slopers in the group that performed intermittent deadhangs compared to the group that just did bouldering.

In turn, López-Rivera and González-Badillo (2019) compared the effects of three different deadhang protocols on endurance. They studied 26 F7c+/8a climbers training for six days a week (not just doing deadhangs).

The MaxHangs group did eight weeks of max-weight deadhangs on a large edge. The training protocol was three to five sets of 10-second hangs (potential hang time of 13 seconds), with three minutes of rest. This protocol tested the effect of maximum strength training on endurance.

The IntHangs group did eight weeks of intermittent deadhangs on an edge just big enough to complete the protocol. The aim was to fail or be close to failing on the last hang of the last set. The climbers could change the size of the edge if necessary. In the first training session, they began on an edge that they could only just hang for 30 seconds. The training protocol was three to five sets of four 10-second hangs, resting for five seconds between hangs and for one minute between sets.

The mixed group (Max\_IntHangs) did four weeks of the MaxHangs protocol and four weeks of the IntHangs protocol.

The test to measure endurance was time to failure on an 11-millimetre edge with no added weight. The results showed that:

Hang-time in the MaxHangs group increased by 34.1 per cent (corroborating everything written in [chapter 2](#) about the benefits of max strength training on endurance).

Hang-time in the IntHangs group increased by an impressive 45 per cent.

Interestingly, endurance in the Max\_IntHangs group increased by only 1 per cent, which is a statistically negligible amount. According to the authors, this surprising result could be due to excessive fatigue from combining both methods in such a short period of time, a possible ceiling effect (this group had the best results in the initial test, so they could have already been closer to their physiological limit) and the fact that this group recorded the most falls, which would have affected the data.

Having seen the positive effects of intermittent deadhangs on endurance, let's look at different training proposals, methods and protocols.

The intermittent deadhang protocol proposed by López-Rivera (2017):

3–5 sets × 4–5 hangs × 10 seconds: 5 seconds/1 minute

In other words: three to five sets of four to five 10-second hangs, resting for five seconds (micro-rest) between each hang and for one minute between sets.

The logic behind this protocol:

Regarding hang time: eight to ten seconds is the time needed for a high-intensity contraction.

Regarding rest between hangs: in crux sections of a route, there's one to five seconds between holds, clipping takes an average of three seconds, and three and five seconds is the amount of time needed for oxidation.

Regarding rest between sets: one minute isn't long enough to recover strength between sets, forcing us to work on endurance.

The intermittent deadhang protocol proposed by Anderson and Anderson (2014):

9 sets  $\times$  6 hangs  $\times$  10 seconds: 5 seconds/3 minutes

This is one of several protocols given in their training guide. It should be done after warming up on the wall for 30 to 40 minutes. Each set should be done on a different hold on the fingerboard. They suggest 10 seconds on/5 seconds off for beginners and 7 seconds on/3 seconds off for more advanced climbers. They also suggest resting for 70 hours between sessions and doing cycles of six to nine sessions before changing to a different type of training.

The intermittent deadhang protocol proposed by Hörst (2016):

2–5 sets  $\times$  3 hangs  $\times$  7 seconds: 53 seconds/5 minutes

This protocol combines maximum strength and endurance training. Add the same amount of weight as for a max-weight 10-second hang, although in the protocol you only hang for seven seconds. Hörst recommends using an edge no smaller than 14 millimetres, and the perceived intensity of each hang should be 9 to 9.5 out of 10; that is, maximum intensity just shy of failure.